

From Formality to Reality: Reassessing Value Added Tax Equity in Dual Economies*

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Abstract

We study the incidence of the value-added tax (VAT) in settings with high informality. While often viewed as regressive, recent literature suggests consumption taxes may be progressive under the crucial assumptions of zero pass-through in informal markets and that low-income households shop predominantly in that sector. We provide product-level evidence on this pass-through assumption by exploiting a temporary VAT exemption in Peru. Using high-frequency price data from formal and informal vendors for food products and a Synthetic Differences-in-Differences strategy, we estimate pass-through of taxes to consumer prices by product and sector. Household survey data confirms that the bottom decile of the income distribution spends 34 percentage points more of their budget in informal markets than the top decile. Our pass-through estimates reveal significant heterogeneity: we find rates of 80-100% in formal markets, but a product-dependent 0-50% in informal markets. These varied rates can be rationalized by a theoretical framework of commodity taxation with endogenous producer prices and heterogeneous household preferences for sectoral amenities. When we combine our empirical findings on shopping behavior and product-level pass-through, the progressivity result disappears, indicating that the VAT remains a regressive instrument in this setting.

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1 Introduction

Indirect taxes imposed on consumption, such as value-added taxes (VAT), are the primary source of government revenue in many developing countries (Jensen, 2022; Brockmeyer, Mascagni, et al., 2024). According to a study by the Organization for Economic Cooperation and Development (OECD, 2022), taxes on goods and services constituted nearly half of the total tax revenue in 26 Latin American Countries (48.4%) in 2021, compared with a third (31.9%) in OECD countries. In contrast, direct taxes such as personal income tax and corporate income tax are not a major source of revenue in developing countries, where a large proportion of the population works in the informal sector and therefore avoids formal tax obligations. Indirect taxes, on the other hand, can be easier to administer and more difficult to evade, as they are embedded in the prices of goods and services and are therefore collected by businesses rather than individuals. This makes indirect taxes an attractive source of revenue for governments that lack the resources to effectively monitor and enforce compliance with tax laws.

In terms of their redistributive impact, uniform consumption taxes are often considered neutral at best, but generally regressive (Lustig, 2018; Warren, 2008) because they impose a larger tax burden on low-income households compared to high-income households. These arguments have led governments to apply a reduced (or zero) rate to necessity goods, such as food items. This is not optimal under the Atkinson and Stiglitz (1976) framework, according to which, under certain assumptions on the utility function¹, it is better to avoid the distortions that arise from differential commodity taxation and redistribute using a nonlinear income tax. However, a key limitation of this framework is that it does not account for informal consumption, defined as purchases that occur outside the formal tax system.

Recent literature (Bachas, Gadenne, and Jensen, 2023) uses store type as a proxy for tax compliance, assuming that modern stores are fully formal and traditional stores are fully informal. This literature also assumes that the rate of tax pass-through to consumer prices is 100% in formal stores and 0% in informal stores. Because low-income households spend a larger share of their budget in traditional (informal) stores, this method leads to the conclusion that uniform consumption taxes can be progressive and reduce inequality in developing countries. However, the assumption of zero tax pass-through in the informal sector is unlikely to hold in reality. Relaxing this assumption is key to understanding the true redistributive impact of consumption taxes in developing countries.

One of the reasons to believe that the pass-through of taxes to prices is greater than zero in informal stores is related to the nature of the VAT structure. Stores typically offset the tax they pay for their purchases with the tax they receive from their sales. If informal stores buy part of their inputs from formal suppliers, the VAT they pay for these purchases is not recovered through fiscal credit because they sell in an informal way. Consequently, the amount of VAT paid for inputs is regarded as an additional cost that will be transferred to consumers. This is part of the well-known formal/informal dualism that makes incidence analysis intricate.

¹These assumptions are weak separability between consumption choices and income in utility, and homogeneity across individuals in the sub-utility of consumption. These two conditions imply that conditional on earnings, consumption choices provide no information on ability, and therefore differentiated commodity taxes create a tax distortion with no benefit.

A further rationale supporting the idea that pass-through might be greater than zero relates to the possibility that informal stores are not entirely informal, remitting taxes for at least a portion of their sales. Finally, competition between informal and formal stores may lead to the former charging similar prices than their formal counterparts, even if they had the same cost structure and they did not remit any of the taxes. As a result, customers at informal stores may end up paying the same or more for identical products instead of paying less, while the tax savings are kept (potentially) entirely by the store.

The assumption that the pass-through to consumer prices is 100% might hold if producers do not bear any of the cost of the tax, but it is not obvious. In fact, Benzarti, Garriga, and Tortarolo (2024) use high-frequency retail scanner data from Argentina to analyze pass-through in different types of stores that would qualify as formal according to the Bachas, Gadenne, and Jensen (2023) framework. They find that a temporary VAT cut (from 21 percent to 0 percent) had a 50 percent pass-through, and its subsequent and equal-sized VAT increase had a 90% pass-through. They also estimate the pass-through of chain and independent supermarkets separately, finding that the pass-through of the VAT cut at the former was 84 percent and the latter 35 percent. They argue that the difference might be due to differences in evasion, competition, or pricing strategies.

It is also important to note that when a final good is exempted from the VAT, taxes paid on inputs are generally not allowed to be used as fiscal credit (this is in contrast to a zero rate tax, in which taxes paid on inputs can be recovered, or in contrast to a law that explicitly specifies that you can recover taxes paid on certain inputs of exempted goods²). Therefore, when a good is exempted, once again in this case, taxes paid on inputs are regarded as costs, and we would expect the price to go down by less than the total amount of the tax, even in the case of perfect compliance.

This paper revisits the question of VAT incidence in countries with substantial informality. We first develop a theoretical framework based on Bachas, Gadenne, and Jensen (2023) with endogenous producer prices and heterogeneous household preferences for sectoral amenities to explain why tax pass-through may differ across sectors. Motivated by this framework, we move to the empirical analysis and look at the case of Peru, which provides a convenient case study for numerous reasons. First, it has high levels of informality, with an informal employment rate of 71.6 percent in 2023³, according to International Labour Organization (2024), and with 58 percent of formal private firms having indicated that they faced competition from unregistered or informal firms in 2023, according to a report from the World Bank’s Enterprise Surveys (World Bank, 2024). Secondly, it has high-frequency data on wholesale and retail prices at the level of market-product-variety for large markets that sell most of their transactions (if not all) in an informal way⁴. Third, we have data for a subset of formal supermarkets that allow us to calculate pass-through of the formal sector in Peru. Lastly, a policy announced in mid-April 2022 temporarily exempted certain products of the basic consumption basket from the value added tax (VAT), originally at 18 percent, starting on May 1st, 2022 through July 31st, 2022. The goal of the policy was to limit the inflationary impact generated by

²The latter is the case for the law that regulates the policy we exploit in this study.

³Calculated as the share of informal employment in total employment for Youth and Adults (ages 15+) based on LFS - Permanent Employment Survey.

⁴That is, without remitting consumption taxes to the tax authority.

the international economic crisis. The law exempted the last stage of production only, but stated that a fiscal credit was to be issued for the amount of VAT paid for the main (but not all) inputs (either domestically bought or imported) used in the production of the exempted goods. This setting provides a natural experiment to estimate the pass-through of taxes to prices for the products subject to a tax change as a result of the policy.

The redistributive analysis relies on detailed expenditure microdata from the 2022 National Household Survey of Peru (ENAHU, for its Spanish acronym). One of the key strengths of the survey is that it collects information on the various types of stores where households purchase products, such as supermarkets, specialty stores, and street vendors. By classifying these stores into formal and informal categories, we gain valuable insight into household consumption patterns in both sectors. This categorization enables us to construct a proxy for household consumption from the informal sector, which, together with the estimation of the pass-through of taxes to prices as outlined in the preceding paragraph, will let us conduct a comprehensive analysis of the impact of consumption taxes on inequality, taking into account the prevalence of informality in the economy.

Using a synthetic differences-in-differences strategy, we estimate that the pass-through of taxes to consumer prices in formal stores is between 80 to 100%, while it ranges from 0 to 47% in informal markets, depending on the product. At the same time, households in the bottom 10 percent of the income distribution spend, on average, 34 percentage points more of their income at informal markets than those in the top 10 percent of the income distribution. Taking these pieces of evidence together, the value-added-tax is considered regressive.

One concern that might arise is how credible is the assumption that the markets on which we have data do not pay taxes on the output sold. To address this, we conducted a survey in one third of these markets in Peru to determine the percentage of their sales that are done in an informal way.⁵ We conducted test purchases and asked questions about business practices and payment methods. We found evidence of low transaction records, with less than 2 percent of the transactions issuing a receipt. To the extent that lower transaction records are associated with lower tax compliance (Brockmeyer, Smith, et al., 2019), we can conclude that the markets we have data on are essentially informal.

Related Literature. Our paper contributes to four key strands of the literature. First, we add to the extensive literature on consumption tax pass-through (Carbonnier, 2007; Kosonen, 2015; Gaarder, 2018; Benedek et al., 2020; Benzarti, Carloni, et al., 2020; Benzarti, Garriga, and Tortarolo, 2024; Brusco and Velayudhan, 2025; Fuest, Neumeier, and Stöhlker, 2025). To this body of work, we provide what is, to our knowledge, one of the first pass-through estimates using high-frequency price data from large informal markets in a developing country. Second, our work speaks to the literature on the influence of tax evasion on the conventional conclusions drawn from previous research (Kopczuk et al., 2016; Harju, Kosonen, and Skans, 2018; Slemrod and Velayudhan, 2022; Brockmeyer, Mascagni, et al., 2024). We follow closely the methodology of Bachas, Gadenne, and Jensen (2023) but with two key empirical differences: we analyze a temporary tax reduction rather than an increase, and our data allow for a differential pass-through analysis by product. Third, we contribute to the literature on

⁵We collected information of 420 stands in 42 different markets in Lima, Callao, and the interior of the country.

measuring the informal sector (Restrepo-Echavarria, 2014; Rothenberg et al., 2016; Charman et al., 2017; Liu et al., 2019). We provide measures through our own survey of the extent of informality in traditional markets, measuring receipt issuance directly. Finally, our paper can fit in the literature on the interaction between formal and informal sectors. This literature has traditionally focused on supply-side linkages and segmented supply chains (Bose, 1998; De Paula and Scheinkman, 2010). In contrast, we focus on the demand side, arguing that these markets are not perfectly segmented for all products. We provide a theoretical framework and empirical evidence suggesting that this interaction is micro-founded on consumer preferences for sector-specific amenities.

The remainder of the paper is organized as follows. In Section 2, we develop a theoretical framework to provide intuition on why pass-through in informal stores might not be zero. Section 3 discusses general aspects of the structure of the value-added tax and the institutional framework surrounding it in Peru. Section 4 outlines the details of the data used in this study. Section 5 describes the main findings from our own survey to traditional markets. Section 6 elaborates on our empirical approach and results for pass-through analysis. Section 7 presents the redistributive analysis. Finally, in Section 8, we summarize our findings and conclude.

2 Theoretical framework

The objective of this section is to provide intuition on why pass-through might be non-zero for informal stores. The results we find latter in the study are consistent with the framework provided by Bachas, Gadenne, and Jensen (2023) with heterogeneous households and endogenous producer prices.

Our primary contribution is to provide a behavioral microfoundation for the demand side that endogenously derives consumer choice between sectors. We propose that a household’s choice of where to shop is a trade-off not just between prices, but also between the distinct non-price amenities offered by each sector. On the supply side, our results are consistent with small to no sector sector switching as a response to the tax.

Households. There are J goods and two varieties d , informal ($d = 0$, untaxed) and formal ($d = 1$, taxed).⁶ To isolate this core trade-off, we assume that households enter the market having already allocated a specific budget (y^i) for the consumption of the good j . Their primary decision is therefore where to spend this budget to maximize their utility, so we ignore the j index for simplicity. The quality of the goods could be equal in both varieties, but the non-price amenities offered by each sector are different. The consumer price of the formal variety is equal to the producer price plus the tax, $p_1 = q_1(1 + t)$. The consumer price of the informal variety is equal to the producer price $p_0 = q_0$. There are N heterogeneous households with a continuum of types that choose where to shop balancing a trade-off not just between consumer prices, but also between the distinct non-price amenities offered by each sector. Households are heterogeneous in two key dimensions: their income level, y^i , which

⁶Taxed means in this context that the tax will be remitted to the tax authority, whereas untaxed means that it will not. However, we will allow the tax to affect the producer prices of both varieties.

is drawn from a distribution with CDF $G(\cdot)$; and a preference for the bundle of amenities offered by each sector, A^i . We assume that these preferences are defined as follows:

$$A^i = \psi(y^i) + \nu^i$$

where $\psi(y^i)$ is a systematic component with $\psi'(y^i) > 0$ (as people get richer, their average preference for the bundle of amenities offered by the formal sector increases) and ν^i an idiosyncratic component that follows a logit distribution with CDF $F(\cdot)$ and a mean of μ , which guarantees that there are consumers of both varieties (variation in personal taste among people with the same income).

Both sectors offer amenities, though of different kinds. Formal stores provide conveniences such as accessible parking, cleaner facilities, air conditioning, credit card payment options, and less crowded spaces—amenities generally preferred by wealthier customers. In contrast, informal stores may offer greater proximity, flexibility with bulk splitting, allowing customers to purchase in smaller quantities—e.g., individual eggs, or single-use packets of detergent—instead of full-sized packages, and often provide an informal credit or tab system for trusted customers. These features make shopping more accessible and manageable for lower-income customers with limited cash flow. Positive values of A^i mean that household i , when deciding where to buy the good, prefers the amenities of the formal sector, negative values indicate a negative preference for the formal amenities, or equivalently a positive valuation of informal amenities. Zero indicates indifference.

This framework allows us to derive demand for each sector in a setting where we can analyze how the willingness of consumers to substitute between them—the cross-price elasticity of demand—depends on the distribution of preferences for these amenities.

Household i choose z^i to maximize utility⁷:

$$U^i = [u(x_1^i) + A^i]d^i + u(x_0^i)(1 - d^i)$$

subject to the following budget constraint:

$$p_0 x_0^i (1 - d^i) + p_1 x_1^i d^i \leq y^i$$

If they choose the formal sector ($d^i = 1$), their consumption is $x_1^i = y^i/p_1$. Their utility is: $V_1 = u(y^i/p_1) + A^i$.

If they choose the informal sector ($d^i = 0$), their consumption is $x_0^i = y^i/p_0$. Their total utility is: $V_0 = u(y^i/p_0)$

The household will choose the formal sector if and only if:

⁷We treat the informal sector amenities as a baseline so that A_j^i represents the extra utility (or disutility) that a household i gets from the formal sector amenities.

$$\begin{aligned}
V_{j1} &\geq V_{j0} \\
u\left(\frac{y_j^i}{p_{j1}}\right) + A_j^i &\geq u\left(\frac{y_j^i}{p_{j0}}\right) \\
u\left(\frac{y_j^i}{q_{j1}(1+t_j)}\right) + A_j^i &\geq u\left(\frac{y_j^i}{q_{j0}}\right) \\
A_j^i &\geq u\left(\frac{y_j^i}{q_{j0}}\right) - u\left(\frac{y_j^i}{q_{j1}(1+t_j)}\right)
\end{aligned}$$

Assuming logarithmic utility:

$$\begin{aligned}
A_j^i &\geq \ln\left(\frac{y_j^i}{q_{j0}}\right) - \ln\left(\frac{y_j^i}{q_{j1}(1+t_j)}\right) \\
A_j^i &\geq \ln(y_j^i) - \ln(q_{j0}) - \ln(y_j^i) + \ln(q_{j1}) + \ln(1+t_j) \\
A_j^i &\geq \ln(q_{j1}) - \ln(q_{j0}) + \ln(1+t_j) \equiv A_j^*
\end{aligned}$$

$$\begin{aligned}
A_j^i \geq A_j^* &\iff \psi(y^i) + \nu_j^i \geq A_j^* \\
&\iff \nu_j^i \geq A_j^* - \psi(y^i) \\
&\iff \nu_j^i \geq \ln(q_{j1}) - \ln(q_{j0}) + \ln(1+t_j) - \psi(y^i)
\end{aligned}$$

Therefore, the share of households that consume the informal variety is given by $F(A_j^* - \psi(y^i))$ (i.e., $N^1 = F(A_j^* - \psi(y^i)) \cdot N$ and the aggregate demand for the informal variety of j will be:

$$\begin{aligned}
D_1^{\text{Total}} &= \int_i \int_\nu x_1^i \cdot 1\{\nu_i > A^* - f(y^i)\} dF(\nu) dG(y) \\
D_1^{\text{Total}} &= \int_i x_1^i \left[1 - F(A^* - f(y^i))\right] \cdot N dG(y)
\end{aligned}$$

$$\begin{aligned}
D_0^{\text{Total}} &= \int_i \int_\nu x_1^i \cdot 1\{\nu_i \leq A^* - f(y^i)\} dF(\nu) dG(y) \\
D_0^{\text{Total}} &= \int_i x_1^i \cdot F(A^* - f(y^i)) \cdot N dG(y)
\end{aligned}$$

Define the uncompensated price elasticities of demand for each variety with respect to each price (for a single income group)⁸ as follows:

$$\begin{aligned}\epsilon_{j11} &= \frac{\partial D_{j1}}{\partial p_{j1}} \cdot \frac{p_{j1}}{D_{j1}} \\ \epsilon_{j00} &= \frac{\partial D_{j0}}{\partial p_{j0}} \cdot \frac{p_{j0}}{D_{j0}} \\ \epsilon_{j10} &= \frac{\partial D_{j1}}{\partial p_{j0}} \cdot \frac{p_{j0}}{D_{j1}} \\ \epsilon_{j01} &= \frac{\partial D_{j0}}{\partial p_{j1}} \cdot \frac{p_{j1}}{D_{j0}}\end{aligned}$$

These elasticities will be a function of the distribution of preferences for amenities, which depend on the income.

This framework provides a direct behavioral foundation for the downward sloping Informality Engel Curve, as the probability of a household choosing the formal sector is an increasing function of its income. With the logarithmic utility assumption, this result is not driven by the direct channel of consumption levels, since the amenity threshold A_j^* is the same for everyone. Instead, the model's income dependence operates entirely through our central assumption about amenity preferences, where the distribution of these preferences is centered differently for each income group.

Notice that, as a consequence of binary choice setup and log utility, the following is true:

$$\begin{aligned}\epsilon_{j11} &= -\epsilon_{j10} \\ \epsilon_{j00} &= -\epsilon_{j01}\end{aligned}$$

The consumer's decision to switch sectors depends only on the price ratio p_{j1}/p_{j0} . Therefore, a 1% increase in the formal price has exactly the same magnitude of impact on the choice as a 1% decrease in the informal price.

Producers. We assume the economy is populated by a continuum of firms, indexed by k , each endowed with a specific productivity level θ^k . Each firm produces a good j , exogenously given. As each firm only produces one good, we will only index firm by k and not kj . Each firm is exogenously designated to a sector, formal ($d^k = 1$) or informal ($d^k = 0$), determined by their underlying productivity level⁹. Output is determined by ability and endogenous effort e^k such that $x^k = e^k \theta^k$. Given their sector, each firm chooses its output level to maximize profits.

⁸The overall market elasticity is the weighted average of the elasticities of each income group, where the weight of a specific group is its share of the total demand.

⁹There is a productivity threshold θ^* under which firms will be informal and over which they will be formal. This is consistent with De Paula and Scheinkman, 2010, and Ulyssea, 2018, where firms can only be informal if they remain small enough (that is, until probability of detection goes to infinity). It also aligns with the intuition that only high-productivity firms are able to overcome the costs of formality (such as taxes and regulations) and have the potential to grow large.

A firm k in the informal sector ($d^k = 0$) chooses its output x_0^k to solve:

$$\max_{x_0^k} \pi_0^k = q_0 x_0^k - c\left(\frac{x_0^k}{\theta^k}\right) - R(x_0^k) \quad (1)$$

where q_0 is the producer price, $c(e^k)$ is a convex cost of effort, and $R(x_0^k)$ is an expected cost of informality. Notice that in our framework informal firms can choose freely their x because their sector designation is fixed. Their low productivity level, together with the expected cost of informality $R(x_0^k)$, with $R'(x_0^k) > 0$; $R''(x_0^k) > 0$;, will act as a natural brake on the firm's size. This yields the individual supply function $x_0^k(q_0; \theta^k)$.

Similarly, a firm k in the formal sector ($d^k = 1$) chooses its output x_1^k to solve:

$$\max_{x_1^k} \pi_1^k = q_1 x_1^k - c\left(\frac{x_1^k}{\theta^k}\right) \quad (2)$$

This yields the individual formal supply function $x_1^k(q_1; \theta^k)$.

First order condition for profit maximization for firms in the informal sector will be:

$$q_0 = c'\left(\frac{x_0^{k*}}{\theta^k}\right) \frac{1}{\theta^k} + R'(x_0^{k*})$$

For the formal sector:

$$q_1 = c'\left(\frac{x_1^{k*}}{\theta^k}\right) \frac{1}{\theta^k} \iff q_1 \theta^k = c'\left(\frac{x_1^{k*}}{\theta^k}\right)$$

Both formal and informal firms choose their output level x_d^{k*} to maximize profits by producing where marginal revenue equals marginal cost. However, the composition of the marginal cost is fundamentally different for the two sectors. A formal firm's marginal cost consists solely of its marginal cost of production. It will therefore produce up to the point where the producer price, q_1 , equals this marginal production cost. In contrast, an informal firm faces an additional cost: the increasing risk of discovery as it grows larger. Its total marginal cost is the sum of its marginal cost of production and this marginal cost of risk ($R'(x)$). An informal firm will therefore produce up to the point where the producer price, q_0 , equals this higher, total marginal cost. Informal firms are typically smaller than their formal counterparts not only because of their lower productivity but also because of the risk of being caught.

Aggregate quantities supplied in each sector are:

$$S_0(q_0) = \int_{k \text{ s.t. } d^k=0} x_0^k(q_0; \theta^k) dk \quad (3)$$

$$S_1(q_1) = \int_{k \text{ s.t. } d^k=1} x_1^k(q_1; \theta^k) dk \quad (4)$$

Since we assume firms do not switch sectors, each firm's optimal output level does not depend on the price in the other sector.

The own-price supply elasticities, ϵ_{j1}^S and ϵ_{j0}^S , are positive, reflecting upward-sloping supply curves. As a direct consequence of our “no switching” assumption, the cross-price supply elasticities are zero: $\epsilon_{j10}^S = \epsilon_{j01}^S = 0$.

$$\epsilon_{j11}^S = \frac{\partial S_{j1}}{\partial q_{j1}} \cdot \frac{q_{j1}}{S_{j1}} \quad (5)$$

$$\epsilon_{j00}^S = \frac{\partial S_{j0}}{\partial q_{j0}} \cdot \frac{q_{j0}}{S_{j0}} \quad (6)$$

Equilibrium. Market clearing sets the total quantity demanded equal to the total quantity supplied in each sector.

$$S_0(q_0) = D_0^{\text{Total}} \quad (7)$$

$$S_1(q_1) = D_1^{\text{Total}} \quad (8)$$

These market clearing conditions define the pass-through of taxes to producer prices.

Pass-through of taxes to producer prices equations become:

$$(\epsilon_{j1}^S - \epsilon_{j1}^D) \gamma_{j1} = \epsilon_{j1}^D + \epsilon_{j10}^D \gamma_{j0} (1 + t_j) \quad (9)$$

$$(\epsilon_{j0}^S - \epsilon_{j0}^D) \gamma_{j0} (1 + t_j) = \epsilon_{j01}^D + \epsilon_{j01}^D \gamma_{j1} \quad (10)$$

Replacing the second equation into the first,

$$\gamma_{j1} = \frac{\epsilon_{j1}^D + \epsilon_{j10}^D \frac{\epsilon_{j01}^D}{(\epsilon_{j0}^S - \epsilon_{j0}^D)}}{\left[\epsilon_{j1}^S - \epsilon_{j1}^D - \epsilon_{j10}^D \frac{\epsilon_{j01}^D}{(\epsilon_{j0}^S - \epsilon_{j0}^D)} \right]} \quad (11)$$

Under the assumption that own price elasticities are stronger than cross-price elasticities, the denominator is positive, and the numerator is negative, leading to $\gamma_j 1 < 0$.

Notice that for informal sector, the pass-through to consumer prices is equal to the pass-through to producer prices, γ_{j0} . For the formal sector, the pass-through to consumer prices is $1 + \gamma_{j1}$.

So, for a reduction in the tax, formal sector producer price will change by γ_{j1} (most likely going up by less than 1), and consumer price will change by $1 + \gamma_{j1}$, most likely going down by less than 1.

Pass-through in informal markets:

$$\gamma_{j0} = \frac{\epsilon_{j01}^D (1 + \gamma_{j1})}{(\epsilon_{j0}^S - \epsilon_{j0}^D)(1 + t_j)} \quad (12)$$

We get that γ_{j0} is unambiguously positive in our model with no sector switching and with $0 \leq (1 + \gamma_{j1}) \leq 1$. When the tax is reduced, price in the informal sector will go down.

3 Institutional setting

Value-added taxes (VATs) are taxes on goods or services levied at multiple stages of production, with the crucial feature of taxes on inputs being credited against taxes on output. They are ultimately borne by the final purchasers (International Monetary Fund, 2014)¹⁰ and offer the advantage of securing revenue by the fact that they are collected throughout the process of production (unlike a retail sales tax) but without distorting production decisions (as a turnover tax does). In the absence of restrictions regarding crediting of taxes that are paid on goods and services used as inputs to production (including those paid on investment goods), a VAT would be economically equivalent to a pure retail sales tax. However, practical considerations, such as restrictions on crediting tax for investment goods and monitoring costs, make it challenging to maintain such equivalence.¹¹

In Peru, our country of study, indirect taxes represented 60.1 percent of total tax revenue in 2021, with taxes on goods and services representing 48.4 percent of total revenue, 41.9 percent coming from general taxes and 6.5 percent from excise taxes (International Monetary Fund, 2023). Peru has a single rate of 18 percent for its VAT, with 2 percentage points corresponding to a tax to the local authorities¹². It is applied to all activities (with the exception of exports and certain products and operations that are exempted), such as buying and selling goods in the country, services, construction contracts, the first sale of real estate made by constructors, and imports of goods. It is collected through the invoice credit method,

¹⁰The real burden of the tax may not fall entirely on consumers but may in part be passed back to suppliers of factors through lower prices received by producers. We are going to set aside this possibility for the purpose of this paper.

¹¹See Brockmeyer, Mascagni, et al. (2024) for a detailed description of how VAT works and its substantial importance in developing countries.

¹²Called “Impuesto de promocion municipal”.

where each seller adds the applicable tax rate to each sale and issues an invoice indicating the tax charged. The buyer, if subject to VAT on their own sales, can offset the input tax paid on their purchases against the output tax charged on their sales. Any remaining tax balance is paid to the authorities, and excess credits can be claimed as a refund. The advantage of this method is that it is robust to the omission from taxation of any intermediate transaction. In cases where a vendor fails to levy or report a tax on a sale, any loss of tax revenue will be precisely compensated if the purchaser also omits to request the credit.

Exemptions in Peru are defined in terms of particular commodities or particular regions. The rationale for exemption in general is related to ameliorating the distributional consequences of the tax, both through the effect on prices that consumers face and through the effect on income, if we believe that producers of the exempted goods are on the lower end of the income distribution (and that part of the tax might be borne by producers through imperfect input credit).

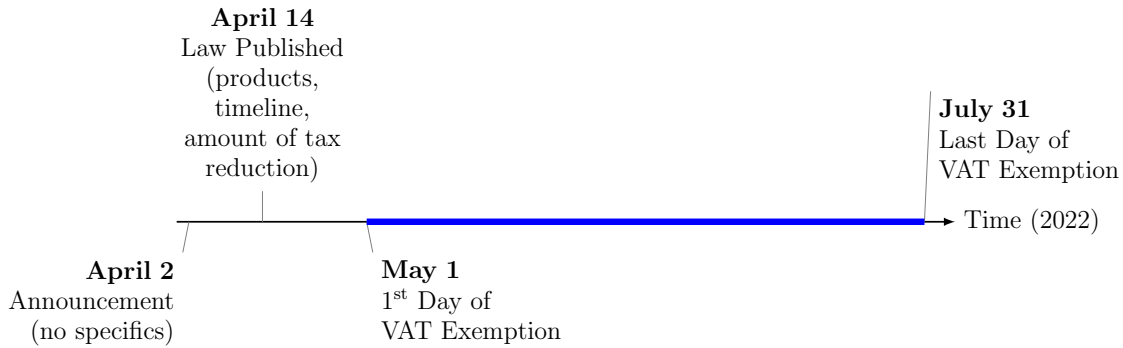
A considerable number of goods and services are exempt in Peru by the VAT law, including the sale or import of certain livestock, fresh vegetables, cereals, phosphates, fertilizers, etc. (see the third column of Appendix Table A1 for a nonexhaustive list), as well as public transport services, cargo transportation, cultural performances, among others that are not relevant for the purpose of this study. These VAT exemptions do not give right of input VAT deduction, meaning that while the tax is not charged on output, tax paid on inputs cannot be reclaimed. Therefore, production decisions (and prices) may be affected by the VAT.

At the same time, the Amazon region has been granted a VAT exemption since 1999, with the enactment of Law 27,037, as part of a government effort to promote development in the area. Despite the fact that this is not a zero-rating arrangement, as no credit is extended for taxes paid on inputs, it can be inferred that since the whole region is exempt, taxes on inputs are not levied. Consequently, this arrangement can be viewed as functionally similar to a zero-rating on output scheme, with consumption taxes exerting minimal influence on production decisions and prices.

The identifying policy variation we exploit is introduced by Law 31,452, which was issued on April 14, 2022.¹³ This law exempted specific food items that are part of the basic consumption basket from the VAT, both when coming from domestic or imported sources, with the aim of mitigating the inflationary impact generated by the ongoing global economic downturn. The exemption was in effect for three months, starting on May 1, 2022, through July 31, 2022, as shown in Figure ?? . Exempted goods comprised fresh, chilled, or frozen poultry meat (specifically *gallus domesticus*), fresh eggs, granulated sugar, uncooked and unstuffed noodles, and bread. The reason behind including these goods was to alleviate the burden of rising food prices on low-income households, who typically spend a higher proportion of their income on these essential goods. At the same time, these are domestically produced goods that contribute significantly to the agricultural sector of the Peruvian economy.

The policy under consideration established that a fiscal credit was to be issued for the amount of VAT paid for the primary inputs (but not all inputs) used in the production of the exempted

¹³The government announced on April 2, 2022 that this law was going to be proposed, although there was no certainty then about the products to be included or the timeline of the policy.



goods, whether being bought from domestic sources or from abroad.¹⁴ This arrangement constitutes a hybrid scenario that lies between exemption and zero-rating. Consequently, taxes will continue to affect production decisions and prices of these goods to some extent. In other words, pass-through could be anywhere between 0 and 100 percent for both formal and informal vendors.¹⁵

The aforementioned policy offers a valuable opportunity to assess pass-through of taxes to prices in both the formal and informal sectors, and this is part of our contribution.

4 Data

4.1 Prices

Formal sector. We have high-frequency data on prices for 80 varieties of products (including 3 of the 5 exempted by the policy¹⁶ sold by 76 supermarkets in Peru in 2022, the year of the policy, on the market level.

Informal sector. This dataset comes from the “Domestic trade statistics of agricultural and agro-industrial food products” project, provided upon request by the Ministry of Agricultural Development and Irrigation of Peru, and elaborated by *Area de comercializacion de la Unidad Estadistica, Oficina de Estudios Economicos y Estadisticos (OEEE)* and *Direcciones de Informacion Agraria Regionales y Subregionales*. It is collected with the goal of keeping agricultural producers and economic agents within the sector informed about the main aspects of the market. It provides high-frequency data (depending on the geographic region and the type of market, as described below) for a total of 330 varieties of products at the market level, including 4 out of the 5 products that were provisionally exempted from the

¹⁴Columns 1 and 2 in Appendix Table A1 show the list of final products and inputs eligible for exemption for which we have data on prices.

¹⁵Potentially pass-through could be greater than 100 if stores over react to the change in tax because of any reason (such as trying to get a higher mark up if they have certain market power by, for example, being the only store in the area). If we call “pass-through” exclusively the part of the price that depends on compliance with the tax, then it is ok to say that pass-through will be bounded by $[0, 100]$. Since in this paper we explore other reasons why the pass-through is not zero, maybe pass-through is not the right word. However, we use this word because we find it to be the most elusive word for the concept we are trying to picture and the most commonly used in the literature.

¹⁶Included: chicken, eggs, and sugar. Excluded: bread and noodles.

VAT under the policy¹⁷. It covers 61 wholesale markets and 67 retail markets throughout the nation, including the metropolitan area of Lima, 23 regional capital cities, and 3 subregional capital cities in Peru.¹⁸ To promote comparability over time and across markets, the people in charge of collecting price information make sure that the product they choose meets the highest quality standards. They collect a sample of three to four prices for each product and report the average in the data.¹⁹ Data for wholesale markets in Lima is reported on a daily basis, while the corresponding data for the interior of the country are available on Mondays, Wednesdays, and Fridays. Data for the retail markets are available on Mondays, Wednesdays, and Fridays for both Lima and the interior of the country.

Some of these markets are publicly owned and the entrance of the sellers is governed by auctions, whereas others are privately owned and allot spaces for rent or sale to vendors. Although wholesale markets are open to both commercial and retail customers, they are typically located farther from urban centers, and consumers generally rely on nearby retail markets for their daily purchases. This paper restricts the analysis to retail markets only, for comparability with the formal sector, although it allows non-treated products in wholesale market to be in the control group.

4.2 Other data

Own data collection. The key premise underlying our analysis is that most transactions in these markets are conducted outside the formal tax system. Even in the cases in which markets are publicly owned and have to comply with tax regulations, it is arguably safe to assume that there is a lot of space for informal transactions to occur, given the government’s limited capacity for regular monitoring and enforcement. To test the accuracy of this assumption, we conducted a survey in May 2024 in 42 markets in our data in Peru (30 in Lima and Callao, 8 in Huánuco, and 4 in Ucayali) asking questions about the types of products offered, price-setting strategies, sales volume, the level of competition within the market and with formal stores, the extent to which clients pay in cash, credit card or other available electronic payment applications²⁰, and other characteristics of the stand. Most importantly, we asked interviewers to register if there were any clients arriving while the survey was conducted, as well as if they had been handed a receipt. Additionally, we randomly selected stands in each market to purchase merchandise from (for a random amount between 5 and 10 soles, enough for these stands to be forced to give you a receipt according to the law) and registered whether a receipt was given. We ran 420 surveys (10 in each market) and made purchases in 311 of the stands.

National Household Expenditure Survey. We use data for consumption patterns across different types of stores from the 2022 National Household Expenditure Survey (in Spanish,

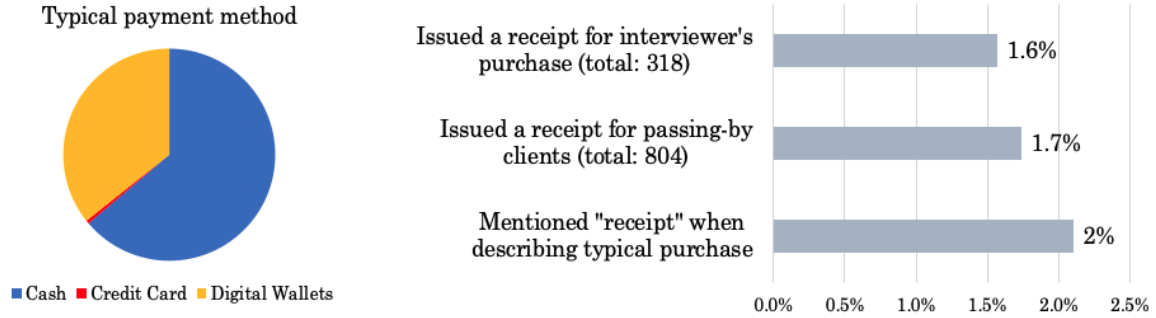
¹⁷Included: chicken, eggs, noodles, and sugar. Excluded: bread.

¹⁸The regional capital cities are Abancay, Arequipa, Ayacucho, Cajamarca, Cerro de Pasco, Chachapoyas, Chiclayo, Cusco, Huaraz, Huancavelica, Huancayo, Huanuco, Ica, Iquitos, Madre de Dios, Moqueagua, Piura, Pucallpa, Puno, Tacna, Tarapoto, Trujillo, and Tumbes. The subregional capital cities are Andahuaylas (in Apurimac), Jaen, and Chota (in Cajamarca).

¹⁹The stands within a market they collect the data from are mobile, meaning that they are not always the same each day.

²⁰The most popular applications of these kind in Peru are called *Yape* and *Plin*. These are comparable to *Venmo* in the US but with a wider acceptance as a payment method in stores.

Figure 1: Survey to traditional markets: summary of results



Source: own elaboration based on survey to traditional markets. Date of collection: May 2024.

“ENAH0”), which is conducted by the National Institute of Informatics and Statistics (in Spanish, “INEI”) which has an expenditure module with a question that explicitly asks in which type of stores you buy each product, with information for 7,550 food products in the second quarter of 2022, the module we use. We classify each purchase as “formal” and “informal” based on the type of store each item is purchased from, with traditional stores and the markets we have data on classified as informal, and modern stores as formal (see Appendix B for details on which store is classified in each group on Table B1, as well as a set of photographs that illustrate some types of stores/vendors). This gives us a proxy for food consumption from the informal sector at the household level, which we use to construct measures of progressivity of the VAT. In particular, we look at the share of food expenditure across the income distribution, differentiating between the store types in which they are bought from, as well as whether the products are subject to the VAT cut or not.

5 Results from Survey to Informal Markets

Figure 1 and Appendix C show a summary of the findings of our survey of informal markets. We find that, out of the 318 purchases made, a receipt was given in only 5 of them (1.6%). Regarding third-party purchases made during the time the survey was conducted, we found that in only 14 of the 804 purchases a client received a receipt (1.7%). We also find that only 1% of purchases are typically paid with credit card, suggesting that there are low transaction records for this type of stores.²¹

Looking at the results of the survey, as well as the receipts given in these markets when requested (as in Figures B4 and B5 in the Appendix B), we can draw several conclusions. First, to the extent that not giving a receipt is equivalent to remitting taxes to the tax authority, around 98.3% of the transactions in these markets are not reported to the tax authority. Secondly, the price we observe in our data includes VAT. This is drawn based on the fact that when clients are given a receipt, the final price does not change (it is the same as the posted price at the markets). Third, people usually do not receive a receipt unless you ask for one. The receipt number in Figure B4 is 371, and the start date of these receipts that we can identify thanks to this tax payer id that we see (RUC - Registro unico de Contribuyente)

²¹The survey questionnaire is available upon request.

is October 10, 2022. It is hard to believe that there were only 371 sales in almost 14 months. Similarly, the one in Figure B5 has 339 registered sales in almost 19 months. This is anecdotal evidence that supports our survey results that even though these stands are registered with the tax authorities, most of their sales are informal, keeping the VAT portion of the prices (or at least the portion corresponding to the sale to the final consumer) to themselves.

6 Pass-through Analysis

6.1 Empirical Strategy

Our main approach to calculate the pass-through of taxes to prices is one in which we restrict the analysis to regions that are always subject to the VAT (non-Amazon), which include the most reliable data for the informal sector and also include formal sector data. ²²

Panel construction. We construct a panel of prices at the market-product level with a weekly frequency, where we keep the first non-missing observation of that week as the data point for that week. We make sure the units of measurement are the same across products in different markets. If they differ (which is rarely the case), we convert that data point to make sure there is homogeneity in the units of measure. ²³

Event Study Design. A natural approach would be to do an event study design as follows, separately for formal and informal markets:

$$\ln p_{mpt} = \sum_{t \neq -1} \beta_t^F \text{week}_t + \sum_{t \neq -1} \gamma_t^F \text{week}_t \cdot \text{treated}_p + \alpha_{mp} + \varepsilon_{mpt} \quad (13)$$

where p_{mpt} is the price of product p in market m in week t , β_t^F reflects the evolution of the price of the control group relative to the week before the policy was implemented, $\beta_t^F + \gamma_t^F$ is the equivalent for treated products in formal markets, α_{mp} are market-product level fixed effects which control for all time-invariant differences between specific products at specific markets. Therefore, our coefficient of interest γ_t^F reflects the estimated difference-in-differences effect for formal markets at week t . It measures the evolution of prices for treated products relative to the evolution of prices for control products, after differencing out the baseline time trend and time-invariant product-market characteristics.

This model would provide a causal estimate of γ_t^F (and similarly, of γ_t^I) under the assumption of parallel trends: that is, the unweighted average of our control products provides a valid counterfactual for the treated products. As reflected in an example in Appendix XX, it is

²² An alternative approach that we explored was to exploit the regional variation of the tax. We restricted the analysis to those products subject to the policy change and compare their prices in regions that are subject to VAT (non-Amazon: treated) with those in regions that are never subject to VAT (Amazon: control). However, the quality of the data we have for the Amazon region is not as good as in the non-Amazon region, resulting in noisier estimates and making it hard to identify a reliable measure of pass-through.

²³ This is why we take the price level comparisons with a grain of salt, since products that come in larger packages will very likely have a lower price per unit.

challenging to achieve this by manually selecting a control group. That is why we rely on a different strategy to calculate pass-through, as described in the next subsection.

Synthetic Differences-in-Differences (SDID). To relax the strong parallel trends assumption required by the standard event study, we employ the Synthetic Difference-in-Differences (SDID) method (Arkhangelsky et al., 2021). This approach combines attractive features of both Differences-in-Differences (like invariance to additive unit-level shifts) and Synthetic controls (reweighting and matching pre-exposure trends to weaken the reliance on parallel trends assumption). The process consists of two main steps: first, it finds optimal **unit weights** ($\hat{\omega}_{mp}$) for each control product in the “donor pool”. These weights are chosen to construct a “synthetic” control group whose price *trend* matches the pre-treatment trend of the treated product as closely as possible. As Arkhangelsky et al. (2021) note, the inclusion of unit fixed effects in the model means it does not require pre-treatment price *levels* to match, only their trends. Second, the method finds optimal **time weights** ($\hat{\lambda}_t$) for the pre-treatment periods in order to construct a baseline difference to correct for the difference in levels between treated and synthetic control. These weights give more importance to past periods that are more predictive of the post-treatment period, which creates a more stable pre-treatment baseline against which to measure the policy’s effect. These weights are then used in a two-way fixed-effects regression to estimate the average causal effect of exposure, denoted by τ , as follows:

$$(\hat{\tau}^{sdid}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau, \mu, \alpha, \beta} \left\{ \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \hat{\omega}_i^{sdid} \hat{\lambda}_t^{sdid} \right\} \quad (14)$$

where $W_{it} \in 0, 1$ denotes treatment status, i is a market-product in our setting, Y is the log of the price of market-product i in week t , N are the total market-products and T are the total time periods.

We implement an event-study like version of this estimation where we get the coefficients τ_t^F and τ_t^I for each of the treated products separately. As an example, for the case of chicken, τ_t^F will reflect the relative difference between the log price of chicken in the formal sector with respect to the log price of its weighted synthetic counterfactual, relative to the average in the pre-treated period.²⁴ Values close to zero in the pre-treatment coefficients give credibility to the identifying assumption.

In each equation we will have one treated product in many markets (either formal or informal). For the control group, we allow both types of markets (formal and informal, and retail and wholesale within the latter) to be in the donor pool of each treated group. We only make sure that untreated products are part of it. We also restrict the donor pool to certain products that are common in the peruvian consumption basket not affected by the policy and for which we have data in both formal and informal varieties.²⁵

²⁴This is in contrast with a traditional event study that picks one unit of comparison, typically the time period before the event.

²⁵These products are: oil, rice, lentils, potatoes, bananas, beans, apples, onions, tomato, papaya, squash, avocado, pineapple. For robustness checks, we also include fish, beef, and milk in some specifications, but we only have data for those for the informal sector. Results do not change much when we include those. The

Among the five treated products, we focus on chicken and eggs. We cannot use bread because of lack of data in any of the sectors, and noodles because of lack of data for the formal variety. For sugar, we decide to ignore it because it had a sharp increase in the price during the month before the policy due to increase in international price that does not let us find a reasonable control group. The treated period spans three months from May 1, 2022 through July 31, 2022.

6.2 Results

Given that the VAT is 18 percent, full pass-through of the tax cut, assuming that all inputs are allowed to be deducted and therefore it would be equivalent to a 0% tax for three months, would mean a reduction of 15.2 percent in the price. We treat the period between announcement and implementation as treated for the purpose of this graph.

Figure 2 presents the dynamic treatment effect estimates (τ_t^F and τ_t^I) for chicken, with corresponding 95% confidence intervals.²⁶

For the formal sector, the result show a sharp and immediate response. In the week of implementation ($t=0$), the price of chicken drops by approximately 13%, consistent with an initial pass-through of the VAT cut of 85 percent. Over the first month of the policy, the average pass-through for chicken in the formal sector was 80 percent.

The informal sector exhibits a more complex, gradual dynamic. A price decline begins shortly after the policy’s announcement -prior to its formal implementation- and continues downward in the subsequent weeks. The average pass-through from the announcement until one month into the policy was approximately 47 percent. This gradual adjustment contrasts sharply with the immediate response of the formal sector.

After the first two weeks of implementation, the relative price of chicken in both sectors begins to increase, potentially reflecting weakening compliance -a phenomenon reported in the news at the time- or other product-specific market adjustments.

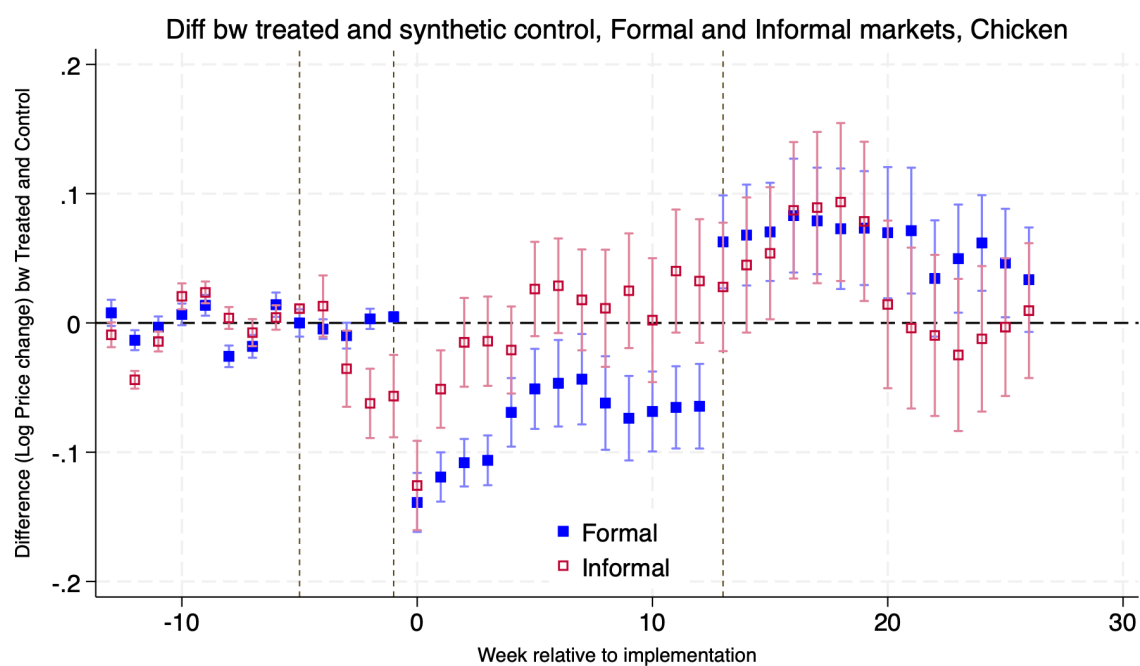
Upon the policy’s termination, the formal sector’s price reverts immediately, with a sharp increase consistent with a 100% pass-through. In contrast, the informal sector’s price adjustment is again more gradual. This asymmetry in response timing suggests that price movements in the informal sector are driven less by direct tax compliance and more by strategic reactions to competitive pressures from the formal market.

Figure 3 shows the corresponding results for eggs. The formal sector again displays a sharp, immediate decline in prices following the policy’s implementation, consistent with a pass-through rate of 100 percent for the first two months of the policy. In stark contrast, the informal sector exhibits a pass-through rate that is statistically indistinguishable from zero for the first six weeks of the policy. If anything, there is a slight increase in the price.

rationale behind this approach is that these products likely have similar elasticity to those included in the policy.

²⁶Confidence intervals are constructed from 50 placebo permutation tests. This non-parametric method is appropriate for a single-treated-unit design and relies on a homoskedasticity assumption, where the variance of the treatment effect is assumed to be similar across units.

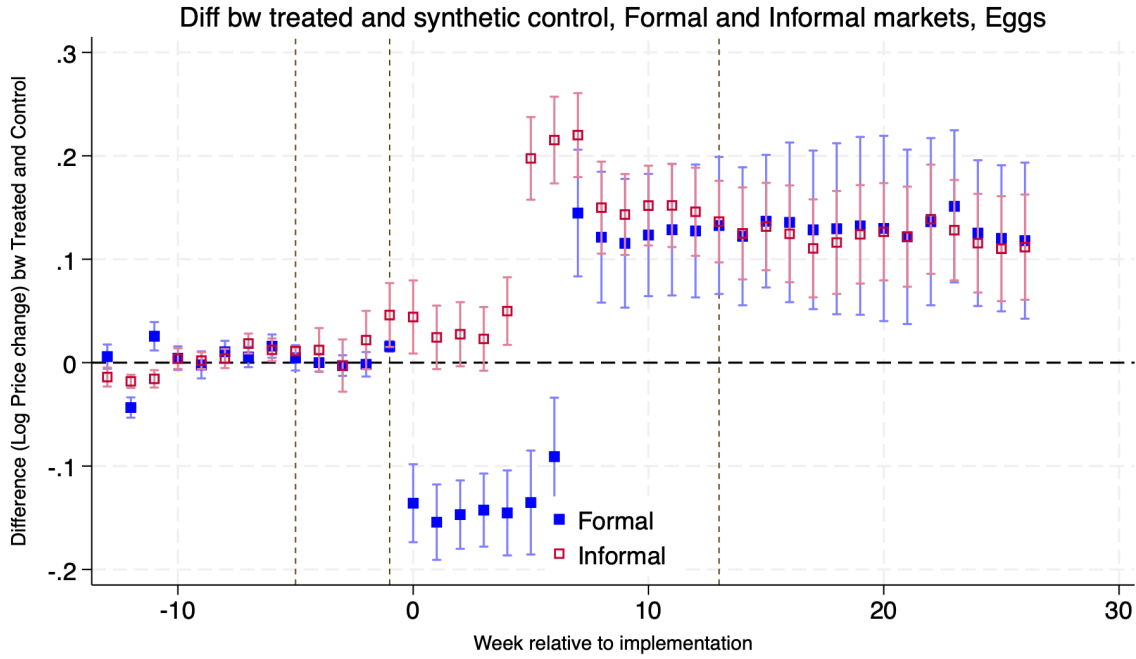
Figure 2: SDID Pass-through estimation for Chicken



Notes: The vertical lines correspond to the week before announcement, implementation, and termination, respectively

Source: own elaboration based on data from the Ministry of Agricultural Development and Irrigation of Peru.

Figure 3: SDID Pass-through estimation for Eggs



Notes: The vertical lines correspond to the week before announcement, implementation, and termination, respectively.

Source: own elaboration based on data from the Ministry of Agricultural Development and Irrigation of Peru.

Notably, both sectors experience a significant price increase around the policy's midpoint. This corresponds to an exogenous shock: a documented rise in the price of corn, a primary input for hen feed. The immediate impact on egg prices is consistent with the production cycle of laying hens, where daily feed costs are a dominant and immediate component of the final product's cost.²⁷

Interpreting the heterogeneity across products. The divergent pass-through rates between chicken (50%) and eggs (0%) in the informal sector can be rationalized by differences in the degree of competition between formal and informal varieties, driven by consumer preferences for sector-specific amenities.

Our results suggest that informal and formal eggs are poor substitutes. While this may result counterintuitive, one possible explanation is that the informal market offers a unique amenity—the ability to purchase eggs by the unit—that is highly valued by consumers with tight daily budgets. This creates a captive customer segment whose preferences are concen-

²⁷While chicken production also relies on corn, the impact of feed price shocks is less immediate. This is due to the shorter, cyclical nature of broiler chicken production, where feed for a given flock is often purchased in advance, and because feed constitutes a relatively smaller share of the final processed product's total value-added compared to eggs.

trated on this feature, making them insensitive to price changes on bulk cartons in supermarkets. In fact, the units sold at supermarkets for which we have data on contained 30 eggs, suggesting that supermarkets tend to offer less options with less units. Insulated from direct competitive pressure, the informal vendor feels no need to pass on the tax cut.

In contrast, informal and formal chicken appear to be close substitutes from the consumer perspective. While the product itself has variety (cuts, brands), the unit of purchase is highly comparable (e.g., per kilogram). Chicken represents a more significant household expenditure, making consumers highly sensitive to price differences on a like-for-like basis. Furthermore, consumers may perceive a quality advantage in the formal sector (e.g., hygiene, refrigeration). Consumer preferences are therefore more dispersed, representing a trade-off between price, convenience, and perceived quality. The informal vendor anticipates losing these “marginal consumers” if the price gap becomes too large, forcing them to react to the supermarket’s price drop by passing on a significant portion of their own cost savings.

7 Redistributive analysis

Regarding the distributional impact of the VAT cut, and considering that the policy was implemented to mitigate the negative impact of the international economic crisis on Peru’s inflation, it is essential to examine household consumption patterns and estimate the share of food expenditure by store type across the income distribution. To achieve this, we use the National Household Survey (ENAHO) data for the second trimester of 2022, a period that coincides with the announcement and first two months of implementation of the policy. In this survey, we calculate the food expenditure for both temporarily exempt products and the rest of the food items (hereby “non-exempt”²⁸), further breaking down the expenditure by different types of stores—formal stores and informal markets, which include those in our data (retail and wholesale markets) and other informal vendors—for each income decile²⁹.

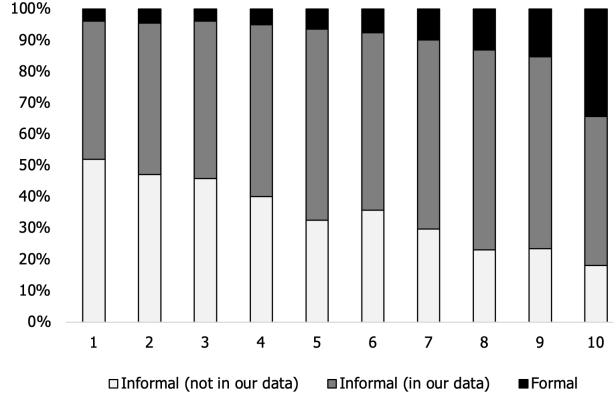
Given the observed differences in pass-through rates between formal and informal stores, we can approximate the tax burden for each store category and, consequently, for households from different income levels. For formal stores, we assign a full pass-through, leading to a 15.2% decrease in final prices for VAT-exempt products. In contrast, the price reduction for informal stores depends on their response to price changes in formal stores. As illustrated in our pass-through estimates, changes in prices in informal stores are product dependent and are consistent with a pass-through of, on average, 25%. Thus, we assume the tax burden for consumers is 15.2% in formal stores and two alternatives for informal stores: 7% as in the case of chicken, and 0% as in the case of eggs, which also serves as a comparison with results in (Bachas, Gadenne, and Jensen, 2023).

Figure 4 illustrate the share of food items purchased in different types of stores by income decile. The data shows that a significant portion of food expenditure occurs in informal stores (two bottom bar categories) across all income levels. This share ranges from 99% for the lowest income decile to 65% for the highest, decreasing as income rises, consistent with findings from previous studies (Bachas, Gadenne, and Jensen, 2023). It should be noted that the difference along the income distribution is even greater for the informal stores that are

²⁸ Although some of these goods include food items that are always exempt (mostly fruits and vegetables).

²⁹ See Table B1 on Appendix B for more details on store types and examples.

Figure 4: Share of Food Expenditure by Store Type and Income Decile



Source: own elaboration based on ENAHO (Apr-Jun, 2022)

in the “Other informal” category, which we believe have an even higher degree of informality than the markets in our data.

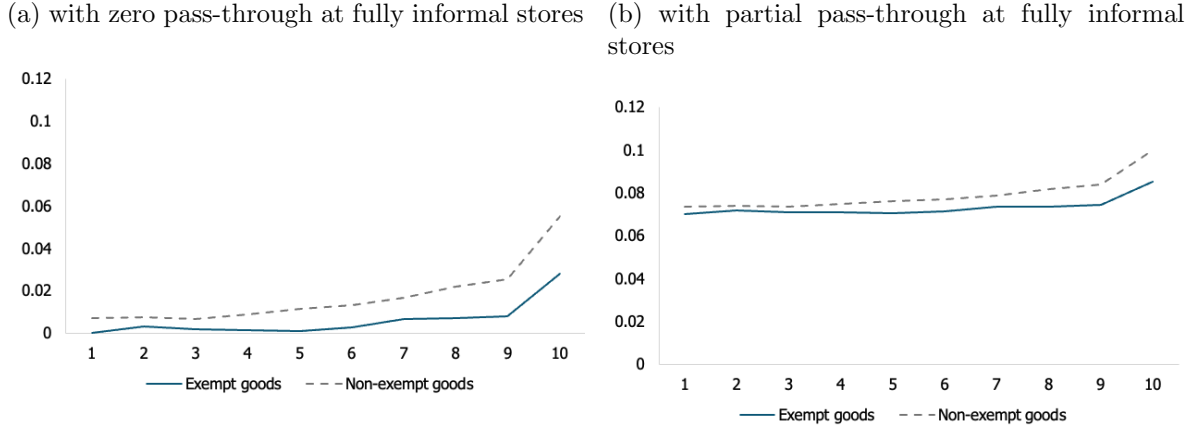
Next we combine these shares and the pass-through results we found before to look at tax liability. We do a back of the envelope approximation of tax liability as the share of taxes that are paid by each decile, taking into account that the tax component in the expenditure we see will be different depending on how much each decile shops at each type of store. As we mentioned before, we assume full pass-through for formal stores, which for a 18% VAT rate corresponds to a 15.2% reduction in final price, and 7% or 0% reduction for informal stores, according to our earlier estimations and for comparison with the literature, respectively. The results assuming zero pass-through for informal stores are presented on Figure 5a, and partial pass-through (7% reduction) on Figure 5b.

In Figure 5a we see that the tax liability for households in the lowest income decile is less than 1%, while it is 3% for the highest decile. In other words, the VAT cut enables poorer households to retain 1% more of their income and wealthier households to retain 3% more. The blue line in Figure 5a illustrates the tax liability for exempted goods across all income deciles, suggesting that the VAT is more progressive than initially assumed once we take informality into account and under certain assumptions of the pass-through.

If we assume that the price response for non-exempted goods mirrors that of exempted goods (as shown in Figure ??), we can estimate the tax liability for non-exempted goods, represented by the dashed line in Figure 5a. The steeper slope of the tax liability curve suggests that the VAT on non-exempted goods is more progressive, meaning richer households bear a higher share of the tax burden. This aligns with the idea that non-exempted goods are less of a necessity and less standardized than exempted goods. Non-exempted goods are more likely to have added value, be sold at (formal) supermarkets, and be purchased by wealthier households.

If instead of assuming zero pass-through for fully informal stores we assume the same pass-through as the partially informal stores that we found in our study for chicken, results change significantly, as shown in Figure 5b. The tax liability now looks flat, and only rises for the highest income decile, suggesting that even when we take into account informality, the

Figure 5: Tax Liability by Income Decile



Source: own elaboration based on data from ENAHO, 2022 April-June.

Notes: tax liability is approximated multiplying the expenditure at each store type by the corresponding pass-through. In both panels, we assume full pass-through for formal stores, which for a 18% VAT rate corresponds to a 15.2% reduction in final price. For informal stores, we show the exercise of 0% pass-through on Figure 5a and partial pass-through (7% reduction) in Figure 5b.

fact that pass-through is not zero at informal stores suggests that the burden of the VAT on household consumption is pretty similar in absolute terms across the income distribution. This is largely because informal stores raise their prices in response to price increases in formal stores due to VAT.

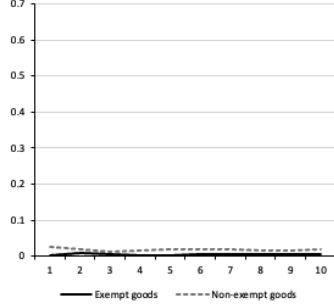
However this exercise by construction would shield a flat curve if we assign the same pass-through through all types of stores. It is also based on deciles of food consumption, which does not differ by a lot in terms of distribution. Perhaps a more informative exercise would be that shown in Figure 6. What we do in this preliminary exercise is, first, assign each decile a proxy of total consumption applying to the total expenditure on food the shares that correspond to income of each decile. Then, we calculate the ratio of tax liability as in the previous exercise, but instead of dividing it by the actual food consumption of that decile, we scale it to mimic the income distribution. I will ignore now the interpretation of the level of this, but the shape is really informative. On Figure 6c we see the classic result of incidence of consumption taxes: if every store charges VAT, the VAT is regressive. In Figure 6a we apply instead a zero pass-through to informal stores, as in Bachas, Gadenne, and Jensen (2023). Because of the fact that in Peru consumption at informal stores is widespread, we obtain a flat curve, getting a neutrality result (instead of progressive result as in Bachas, Gadenne, and Jensen (2023)). If we apply instead the partial pass-through that we find in this paper (as in Figure 6b, we see that the VAT is still regressive.

8 Conclusion

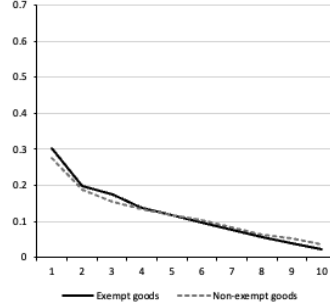
This paper investigates the incidence of the value-added tax in economies with large informal sectors. Recent literature has suggested that consumption taxes may be progressive in such settings, but this result hinges on strong assumptions about tax pass-through. We use a

Figure 6: Tax Liability by Income Decile

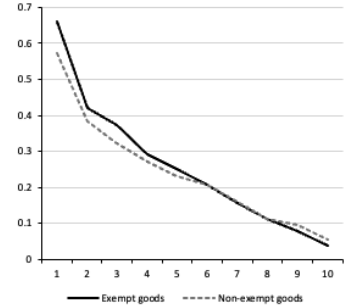
(a) Pass-through: 0% at informal, 100% at formal



(b) Pass-through: 47% at informal, 100% at formal



(c) Pass-through: 100% at formal and informal



Source: own elaboration based on data from ENAHO, 2022 April-June.

Notes: first we rescale total consumption in food assigning each decile the percentile as if it mirrored the income distribution. Then, we proxy tax liability by looking at the share of food expenditure in informal and formal stores, with the corresponding pass-through.

temporary VAT exemption in Peru and a Synthetic Difference-in-Differences design to provide quasi-experimental estimates of pass-through at the product level.

Our findings reveal significant heterogeneity. Pass-through in the formal sector is high, ranging from 80-100%, as expected. In the informal sector, however, we estimate pass-through rates between 0% and 50%. This heterogeneity can be rationalized by the degree of substitutability between formal and informal varieties of a good, which is driven by consumer preferences for sector-specific amenities. When we combine these empirically estimated, non-zero pass-through rates with household survey data on shopping patterns, the previously documented progressivity of the VAT disappears. Our results suggest that policymakers cannot rely on the informal sector to automatically shield low-income households from the burden of consumption taxes, and that a more nuanced, product-level understanding of market structure is essential for accurate incidence analysis.

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Appendices

A Appendix A

Table A1: Product classification into treatment and control groups

Treated	Maybe Treated (Inputs)	Control		
Temporary VAT exemption	Taxes on these inputs are allowed to be used as fiscal credit	Always VAT exempted	Standard 18% VAT (Never exempted)	VAT exemption in Amazon region
Sugar (21 varieties)	Cooking oils (7 varieties)	Garlic	Quinoa	Same products as in Treated and Maybe Treated but in regions that are always exempted from the VAT
Chicken	Flour (3 varieties)	Oats	Rice	
Pasta (5 varieties)	Chicken (poultry)	Onion	Beef	
Eggs		Strawberries	Pork	
		Beans	Squash	
		Milk	Tuna	
		Lettuce	Corn	
		Lemon and limes	Turkey	
		Apples	Basil	
		Avocado	Bell pepper	
		...	...	
[28 product-varieties]	[11 product-varieties]	[176 product-varieties]	[99 product-varieties]	

B Appendix B

Table B1: Classification of store type reported in ENAHO as formal, informal, or other

Formal	Informal	Others
Bakery	"Ambulante" (street vendor/peddler)	Others
Supermarket	"Bodega" (traditional stores)	
Restaurants and bars	Wholesale and retail street markets	
	Foodtrucks/vans	
	Fairs	

Figure B1: "Informal" Markets in Lima, Peru



Figure B2: “Bodega”



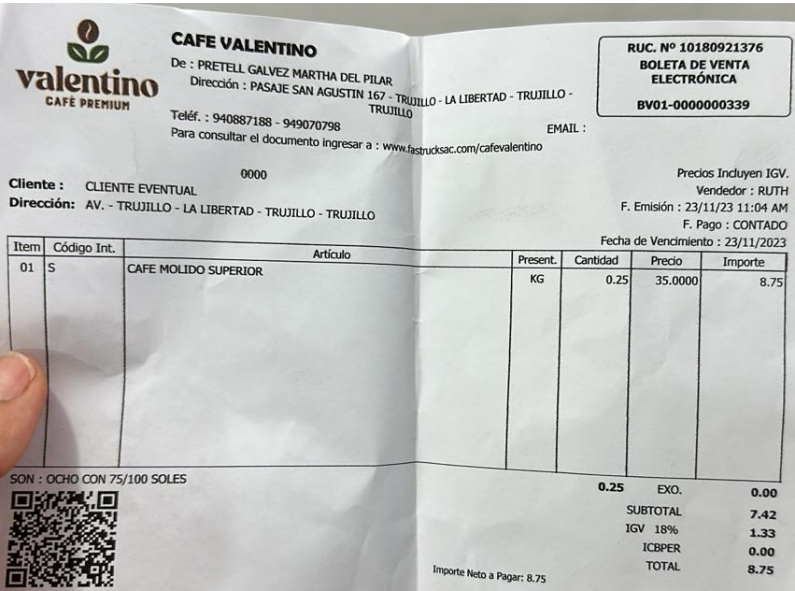
Figure B3: “Ambulante”



Figure B4: “Start date: Oct-10-2022”



Figure B5: “Start date: May-11-2022”



C Appendix C. Own Survey to Informal Markets

Table C1: Survey to Markets. Summary statistics

		All	Type of stand			
			Groceries	Animals and Eggs	Sausages & similar	Fruits and Vegetables
Share surveyed stands		100%	24%	32%	15%	29%
Average number of workers		1.60	1.51	1.81	1.53	1.46
Average number of customers in a day						
	1-15	3%	1%	4%	0%	5%
	16-30	27%	32%	26%	25%	24%
	31-50	38%	33%	43%	39%	37%
	51-100	22%	21%	16%	27%	26%
	101+	11%	14%	11%	9%	7%
Average purchase value (in soles)						
	s/1-5	3%	0%	2%	3%	5%
	s/6-10	23%	17%	18%	20%	33%
	s/11-20	42%	37%	34%	48%	51%
	s/21-40	25%	39%	33%	17%	7%
	s/41-60	4%	7%	4%	6%	1%
	s/61+	4%	0%	8%	5%	2%
Mentioned "receipt" while describing a typical purchase		2%	4%	2%	3%	0%
Typical payment method						
	Cash	64%	64%	64%	60%	66%
	Credit Card	1%	0%	1%	1%	0%
	Yape or Plin	36%	36%	35%	39%	34%
Made a purchase here		74%	84%	71%	80%	67%
% that were given a receipt (without asking for one)		1.6%	4.7%	0.0%	0.0%	1.2%
N other customers buying during survey		804	179	214	149	259
% other customers that were given a receipt		1.7%	2.2%	2.3%	0.0%	0.8%

Table C2: Survey to Markets. Price determination factors, by product

	Relevance of each factor on determining final price (0=nothing, 1=little, 2=relevant, 3=very relevant)																				Level of competition		
	Cost + Margin				Demand				Decide alone, looking at other stands in the market				Decide alone, looking at supermarkets				Decide together with other stands in the market						
Product	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	None	Little	High
apple	0%	28%	26%	47%	88%	9%	2%	0%	93%	7%	0%	0%	84%	12%	5%	0%	36%	36%	21%	7%	0%	30%	70%
avocado	2%	16%	41%	41%	91%	9%	0%	0%	93%	5%	0%	2%	95%	5%	0%	0%	30%	43%	20%	7%	0%	23%	77%
banana	0%	23%	38%	38%	83%	13%	4%	0%	94%	4%	2%	0%	85%	9%	7%	0%	32%	38%	17%	13%	0%	13%	87%
beef	0%	0%	100%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	20%	0%	80%	0%	13%	38%	50%
cheese	0%	21%	44%	35%	90%	7%	2%	2%	91%	5%	3%	0%	83%	16%	2%	0%	31%	43%	7%	19%	3%	21%	76%
chicken	1%	20%	39%	39%	82%	15%	2%	0%	92%	5%	2%	1%	88%	7%	5%	0%	32%	19%	14%	35%	4%	10%	87%
eggs	1%	18%	38%	43%	83%	9%	6%	2%	89%	10%	1%	0%	87%	13%	0%	0%	42%	22%	16%	20%	0%	12%	88%
fish	0%	0%	100%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	0%	17%	83%
honey	0%	0%	100%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	0%	100%	0%
legumes	0%	0%	0%	100%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	0%	0%	50%	50%	0%	0%	100%
noodles	2%	28%	39%	31%	36%	30%	20%	14%	85%	12%	3%	0%	94%	6%	0%	0%	88%	8%	3%	1%	1%	15%	84%
olives	0%	11%	50%	39%	67%	28%	6%	0%	94%	6%	0%	0%	78%	17%	6%	0%	17%	28%	17%	39%	0%	22%	78%
onion	0%	24%	44%	32%	84%	12%	0%	4%	88%	12%	0%	0%	80%	20%	0%	0%	16%	44%	20%	20%	0%	20%	80%
pork	0%	0%	33%	67%	67%	33%	0%	0%	100%	0%	0%	0%	50%	33%	17%	0%	0%	100%	0%	0%	17%	33%	50%
potato	0%	19%	43%	38%	81%	10%	5%	5%	95%	5%	0%	0%	90%	10%	0%	0%	14%	38%	24%	24%	0%	10%	90%
rice	2%	21%	51%	26%	83%	11%	5%	1%	91%	6%	3%	0%	90%	9%	1%	0%	35%	36%	11%	17%	0%	17%	83%
sausage	2%	13%	44%	40%	77%	17%	2%	4%	88%	10%	0%	2%	83%	15%	2%	0%	38%	33%	15%	13%	4%	27%	69%
sugar	1%	47%	28%	24%	87%	3%	10%	0%	93%	0%	7%	0%	93%	1%	6%	0%	35%	12%	36%	17%	2%	22%	76%
tomato	0%	26%	43%	30%	70%	30%	0%	0%	87%	9%	4%	0%	70%	22%	9%	0%	13%	48%	30%	9%	0%	22%	78%
turkey	0%	0%	100%	0%	-	-	-	-	100%	0%	0%	0%	100%	0%	0%	0%	0%	0%	100%	0%	0%	100%	0%
average	1%	24%	40%	35%	78%	13%	6%	3%	91%	6%	2%	0%	88%	10%	3%	0%	39%	28%	17%	16%	2%	19%	80%

treated (2022 policy)